

Lab 2: Hardware and System Information Commands

A: Hardware Information Commands

1. Command: lscpu

Interpretation: Display information about the CPU architecture.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ lscpu  
Architecture:          x86_64  
CPU op-mode(s):        32-bit, 64-bit  
Address sizes:          48 bits physical, 48 bits virtual  
Byte Order:             Little Endian  
CPU(s):                 16  
On-line CPU(s) list:    0-15  
Vendor ID:              AuthenticAMD  
Model name:             AMD Ryzen 7 6800H with Radeon Graphics  
CPU family:             25  
Model:                  68  
Thread(s) per core:     2  
Core(s) per socket:     8  
Socket(s):              1  
Stepping:               1
```

2. Command: lshw

Interpretation: Extract detailed information about the hardware configuration of the machine.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ lshw  
WARNING: you should run this program as super-user.  
nexus  
  description: Computer  
  width: 64 bits  
  capabilities: smp vsyscall32  
*-core  
  description: Motherboard  
  physical id: 0  
*-memory  
  description: System memory  
  physical id: 0  
  size: 7808MiB  
*-cpu  
  product: AMD Ryzen 7 6800H with Radeon Graphics
```

3. Command: sudo lshw

Interpretation: Extract detailed information on the hardware configuration of the machine (requires superuser privileges).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo lshw  
nexus  
  description: Computer  
    width: 64 bits  
    capabilities: smp vsyscall32  
*-core  
  description: Motherboard  
    physical id: 0  
*-memory  
  description: System memory  
    physical id: 0  
    size: 7808MiB  
*-cpu  
  product: AMD Ryzen 7 6800H with Radeon Graphics  
  vendor: Advanced Micro Devices [AMD]
```

4. Command: hwinfo

Interpretation: Probe for the hardware present in the system and generate a system overview log for support.

Output:

```
samir@NEXUS: ~  
===== start debug info =====  
libhd version 21.72u (x86-64) [7688]  
using /var/lib/hardware  
kernel version is 5.15  
----- /proc/cmdline -----  
initrd=\initrd.img WSL_ROOT_INIT=1 panic=-1 nr_cpus=16 hv_utils.timesync_implicit=1 console=hv  
c0 debug pty.legacy_count=0  
----- /proc/cmdline end -----  
debug = 0xffffffffff  
probe = 0x15938fcdaa17fcf9fffe (+memory +pci +isapnp +net +floppy +misc +misc.serial +misc.par +  
misc.floppy +serial +cpu +bios +monitor +mouse +scsi +usb -usb.mods +modem +modem.usb +parallel  
+parallel.lp +parallel.zip -isa -isa.isdn +isdn +kbd +prom +sbus +tint +braille +braille.alva +br  
aille.fhp +braille.ht -ignx11 +sys -bios.vbe -isapnp.old -isapnp.new -isapnp.mod +braille.baum -  
manual +fb +pppoe -scan +pcmcia +fork -parallel.imm +s390 +cpuemu -sysfs -s390disks +udev +block
```

5. Command: `hwinfo --short`

Interpretation: Show a quick summary of the hardware present in the system.

Output:

[illegible]

6. Command: lspci

Interpretation: Display information about PCI buses in the system and devices connected to them.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ lspci  
6afb:00:00.0 SCSI storage controller: Red Hat, Inc. Virtio console (rev 01)  
8306:00:00.0 System peripheral: Red Hat, Inc. Virtio file system (rev 01)  
d844:00:00.0 3D controller: Microsoft Corporation Device 008e  
eb3b:00:00.0 3D controller: Microsoft Corporation Device 008e  
samir@NEXUS:~$ |
```

7. Command: ls SCSI

Interpretation: List SCSI or host devices attached to the system.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ ls SCSI  
[0:0:0:0] disk Msft Virtual Disk 1.0 /dev/sda  
[0:0:0:1] disk Msft Virtual Disk 1.0 /dev/sdb  
[0:0:0:2] disk Msft Virtual Disk 1.0 /dev/sdc  
samir@NEXUS:~$ |
```

8. Command: sudo -s

Interpretation: Run the shell specified by the SHELL environment variable or the shell specified by the invoking user's password database.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo -s  
root@NEXUS:/home/samir# exit  
exit  
samir@NEXUS:~$ █
```

9. Command: inxi

Interpretation: Command line system information script for console and IRC.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ inxi  
CPU: 8-core AMD Ryzen 7 6800H with Radeon Graphics (-MT MCP-) speed: 3194 MHz  
Kernel: 5.15.153.1-microsoft-standard-WSL2 x86_64 Up: 24m  
Mem: 699.6/7564.3 MiB (9.2%) Storage: 1 TiB (22.8% used) Procs: 26  
Shell: Bash inxi: 3.3.13  
samir@NEXUS:~$
```

10. Command: inxi -fx

Interpretation: Show all CPU flags used in short format.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ inxi -fx  
CPU:  
Info: 8-core model: AMD Ryzen 7 6800H with Radeon Graphics bits: 64  
type: MT MCP arch: Zen 3 rev: 1 cache: L1: 512 KiB L2: 4 MiB L3: 16 MiB  
Speed (MHz): avg: 3194 min/max: N/A cores: 1: 3194 2: 3194 3: 3194  
4: 3194 5: 3194 6: 3194 7: 3194 8: 3194 9: 3194 10: 3194 11: 3194 12: 3194  
13: 3194 14: 3194 15: 3194 16: 3194 bogomips: 102208  
Flags: 3dnowprefetch abm adx aes apic arat avx avx2 bmi1 bmi2 clflush  
clflushopt clwb clzero cmov cmp_legacy constant_tsc cpuid cr8_legacy cx16  
cx8 de decodeassists extd_apicid f16c flushbyasid fma fpu fsgsbase fxsr
```

11. Command: inxi -Fx

Interpretation: Show all CPU flags used in detailed format.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ inxi -Fx  
System:  
Host: NEXUS Kernel: 5.15.153.1-microsoft-standard-WSL2 x86_64 bits: 64  
compiler: gcc v: 11.2.0 Desktop: N/A  
Distro: Ubuntu 22.04.5 LTS (Jammy Jellyfish)  
Machine:  
Message: No machine data: try newer kernel. Is dmidecode installed? Try -M  
--dmidecode.  
Battery:  
ID-1: BAT1 charge: 3.0 Wh (60.0%) condition: 5.0/5.0 Wh (100.0%) volts: 5.0  
min: 5.0 model: Microsoft Hyper-V Virtual Batte status: N/A  
CPU:  
Info: 8-core model: AMD Ryzen 7 6800H with Radeon Graphics bits: 64
```

B: Storage and File System Commands

12. Command: lsblk

Interpretation: List information about all available or the specified block devices.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ lsblk  
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS  
sda    8:0    0 388.6M  1 disk  
sdb    8:16    0     2G   0 disk [SWAP]  
sdc    8:32    0     1T   0 disk /mnt/wslg/distro  
/
```

13. Command: df

Interpretation: Display the amount of space available on the file systems.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ df  
Filesystem      1K-blocks      Used    Available    Use% Mounted on  
none            3872940           4      3872936         1% /mnt/wsl  
drivers         440402244 243990712 196411532      56% /usr/lib/wsl/drivers  
none            3872940           0      3872940         0% /usr/lib/modules  
none            3872940           0      3872940         0% /usr/lib/modules/5.15.153.1-m  
icrosoft-standard-WSL2  
/dev/sdc        1055762868 1396268 1000663128      1% /  
none            3872940           84      3872856         1% /mnt/wslg  
none            3872940           0      3872940         0% /usr/lib/wsl/lib  
rootfs          3869588        2080      3867508         1% /init  
none            3872940          528      3872412         1% /run  
none            3872940           0      3872940         0% /run/lock
```

14. Command: pydf

Interpretation: Python script that displays the amount of disk space available on the mounted file systems, using different colors for different types of file systems.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ pydf  
Filesystem  Size  Used Avail Use% Mounted on  
/dev/sdc    1007G 1364M 954G  0.1 [.....] /  
rootfs      3779M 2080k 3777M 0.1 [.....] /init  
C:\134      420G  233G 187G 55.4 [###...] /mnt/c  
D:\134       34G   79M  34G  0.2 [.....] /mnt/d  
/dev/sdc    1007G 1364M 954G  0.1 [.....] /mnt/wslg/distro  
none        3782M 100k 3782M 0.0 [.....] /mnt/wslg/doc  
none        3782M 100k 3782M 0.0 [.....] /mnt/wslg/versions.txt  
none        3782M   0 3782M 0.0 [.....] /usr/lib/modules/5.~soft-standard-WSL2  
drivers     420G  233G 187G 55.4 [###...] /usr/lib/wsl/drivers  
none        3782M   0 3782M 0.0 [.....] /usr/lib/wsl/lib  
samir@NEXUS:~$
```

15. Command: fdisk

Interpretation: Dialog-driven program for creation and manipulation of partition tables. It understands GPT, MBR, Sun, SGI, and BSD partition tables.

16. Command: `sudo fdisk -l`

Interpretation: List the partition table for the specified devices (requires superuser privileges).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo fdisk -l  
Disk /dev/ram0: 64 MiB, 67108864 bytes, 131072 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 4096 bytes  
I/O size (minimum/optimal): 4096 bytes / 4096 bytes  
  
Disk /dev/ram1: 64 MiB, 67108864 bytes, 131072 sectors  
Units: sectors of 1 * 512 = 512 bytes  
Sector size (logical/physical): 512 bytes / 4096 bytes  
I/O size (minimum/optimal): 4096 bytes / 4096 bytes
```

17. Command: `free`

Interpretation: Display the amount of free and used space in the device.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ free  
              total        used        free      shared  buff/cache   available  
Mem:           7745884      452116      6827684        2964       466084      7055328  
Swap:          2097152           0       2097152  
samir@NEXUS:~$ █
```

C: Processor and Memory Information

18. Command: `dmidecode -t processor`

Interpretation: Dump the computer's DMI (SMBIOS) table contents in a human-readable format, specifically the processor information.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ dmidecode -t processor  
# dmidecode 3.3  
Scanning /dev/mem for entry point.  
/dev/mem: Permission denied  
samir@NEXUS:~$ sudo dmidecode -t processor  
# dmidecode 3.3  
Scanning /dev/mem for entry point.  
# No SMBIOS nor DMI entry point found, sorry.  
samir@NEXUS:~$ █
```

19. Command: dmidecode -t memory

Interpretation: Dump the computer's DMI (SMBIOS) table contents, specifically the memory device information.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo dmidecode -t memory  
# dmidecode 3.3  
Scanning /dev/mem for entry point.  
# No SMBIOS nor DMI entry point found, sorry.  
samir@NEXUS:~$
```

20. Command: dmidecode -t bios

Interpretation: Dump the computer's DMI (SMBIOS) table contents, specifically the BIOS information.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo dmidecode -t bios  
# dmidecode 3.3  
Scanning /dev/mem for entry point.  
# No SMBIOS nor DMI entry point found, sorry.  
samir@NEXUS:~$
```

21. Command: cat /proc/cpuinfo

Interpretation: Read the /proc/cpuinfo file to output the processor's information, such as CPU family, model, microcode, flags, address sizes, and cache alignments.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ cat /proc/cpuinfo  
processor       : 0  
vendor_id      : AuthenticAMD  
cpu family     : 25  
model          : 68  
model name     : AMD Ryzen 7 6800H with Radeon Graphics  
stepping       : 1  
microcode      : 0xffffffff  
cpu MHz        : 3194.005  
cache size     : 512 KB  
physical id    : 0  
siblings       : 16  
cpu id         : 0
```

22. Command: cat /proc/meminfo

Interpretation: Read the /proc/meminfo file and output the various information about the memory segments of the computer system.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ cat /proc/meminfo  
MemTotal:       7745884 kB  
MemFree:        6606000 kB  
MemAvailable:   7048668 kB  
Buffers:        12756 kB  
Cached:         625964 kB  
SwapCached:      0 kB  
Active:         101992 kB  
Inactive:       672144 kB  
Active(anon):    2376 kB  
Inactive(anon):  136008 kB  
Active(file):    99616 kB  
Inactive(file):  536136 kB
```

D: System Information Commands

23. Command: cat /proc/version

Interpretation: Display a quick summary of the system information.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ cat /proc/version  
Linux version 5.15.153.1-microsoft-standard-WSL2 (root@941d701f84f1) (gcc (GCC) 1  
1.2.0, GNU ld (GNU Binutils) 2.37) #1 SMP Fri Mar 29 23:14:13 UTC 2024  
samir@NEXUS:~$
```

24. Command: uname

Interpretation: Print the system name.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname  
Linux  
samir@NEXUS:~$
```

25. Command: uname -n

Interpretation: Print the network host name.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname -n  
NEXUS  
samir@NEXUS:~$
```

26. Command: `uname -v`

Interpretation: Print the kernel version.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname -v  
#1 SMP Fri Mar 29 23:14:13 UTC 2024  
samir@NEXUS:~$
```

27. Command: `uname -r`

Interpretation: Print the kernel release.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname -r  
5.15.153.1-microsoft-standard-WSL2  
samir@NEXUS:~$
```

28. Command: `uname -m`

Interpretation: Print the machine hardware name.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname -m  
x86_64  
samir@NEXUS:~$
```

29. Command: `uname -a`

Interpretation: Print all system information, including kernel name, network node hostname, kernel release and version, machine hardware name, and processor type.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ uname -a  
Linux NEXUS 5.15.153.1-microsoft-standard-WSL2 #1 SMP Fri Mar 29 23:14:13 UTC 2024  
x86_64 x86_64 x86_64 GNU/Linux  
samir@NEXUS:~$
```

30. Command: `sudo lshw -html>lshw.html`

Interpretation: Generate a detailed hardware configuration report in HTML format and save it to the file `lshw.html` (requires superuser privileges).

Output:



```
id: nexus  
description: Computer  
width: 64 bits  
capabilities: smp vsyscall32  
  
id: core  
description: Motherboard  
physical id: 0  
  
id: memory  
description: System memory  
physical id: 0  
size: 7808MiB  
  
id: cpu  
product: AMD Ryzen 7 6800H with Radeon Graphics  
vendor: Advanced Micro Devices [AMD]  
physical id: 1  
bus info: cpu@0  
version: 25.68.1  
width: 64 bits  
capabilities: fpu fpu_exception wp vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt pdpe1gb rdtscp x86-64 constant_tsc rep_good nopl tsc_reliable nonstop_tsc cpuid extd_apicid pni pclmulqdq sse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a misalignsse 3dnowprefetch osvw topoext perfctr_core ssbd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi2 invpcid rdseed adx smap clflushopt clwb sha_ni xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat npt nrpt save_tsc_scale vmcb_clean flushbyasid decodeassists pausefilter pthreshold v_vmsave_vmload umip vaes vpcmlqdq rdpid  
configuration: microcode = 4294967295
```

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo lshw -html>lshw.html  
samir@NEXUS:~$ ls  
demo.txt  lshw.html  
samir@NEXUS:~$ cat lshw.html  
<?xml version="1.0" encoding="utf-8"?>  
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Strict//EN"  
  "http://www.w3.org/TR/xhtml1/DTD/xhtml1-strict.dtd">  
<html xmlns="http://www.w3.org/1999/xhtml">  
<head>  
<meta name="generator" content="lshw-" />
```

E: Calculations and User Information

31. Command: `factor 556`

Interpretation: Print the prime factors of the given number (556).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ factor 556  
556: 2 2 139  
samir@NEXUS:~$
```

32. Command: `echo '784594*488448' | bc`

Interpretation: Use the echo command to display the string '784594*488448', then pipe the output to the bc command (arbitrary precision calculator) to calculate the product.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ echo '784594*488448' | bc  
383233370112  
samir@NEXUS:~$
```

33. Command: `echo '((7*5)+(2+3))/8' | bc`

Interpretation: Use the echo command to display the expression '((7*5)+(2+3))/8', then pipe the output to the bc command to evaluate the expression.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ echo '((7*5)+(2+3))/8' | bc  
5  
samir@NEXUS:~$
```

34. Command: `id`

Interpretation: Print the effective user and group IDs.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ id  
uid=1000(samir) gid=1000(samir) groups=1000(samir),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),117(netdev)  
samir@NEXUS:~$
```

F: Network Diagnostics

35. Command: `ping -w 3 google.com`

Interpretation: Send ICMP ECHO_REQUEST packets to the [google.com](https://www.google.com) domain for 3 seconds.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ ping -w 3 google.com  
PING google.com (142.250.192.238) 56(84) bytes of data.  
64 bytes from del11s13-in-f14.1e100.net (142.250.192.238): icmp_seq=1 ttl=113 time=23.1 ms  
64 bytes from del11s13-in-f14.1e100.net (142.250.192.238): icmp_seq=2 ttl=113 time=19.5 ms  
64 bytes from del11s13-in-f14.1e100.net (142.250.192.238): icmp_seq=3 ttl=113 time=21.1 ms  
  
--- google.com ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2000ms  
rtt min/avg/max/mdev = 19.497/21.232/23.123/1.484 ms  
samir@NEXUS:~$
```

36. Command: netstat

Interpretation: Print network connections, routing tables, interface statistics, masquerade connections, and multicast memberships.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ netstat  
Active Internet connections (w/o servers)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 172.28.164.27:43116    ubuntu-mirror-2.ps:http TIME_WAIT  
tcp        0      0 172.28.164.27:44116    carpaccio.canonic:https ESTABLISHED  
  
Active UNIX domain sockets (w/o servers)  
Proto RefCnt Flags               Type                   State                  I-Node   Path  
unix    2      [ ]                 DGRAM                    
unix    2      [ ]                 DGRAM                    
unix    3      [ ]                 DGRAM                  CONNECTED              26681    /run/systemd/notify  
unix    2      [ ]                 DGRAM                    
unix    8      [ ]                 DGRAM                  CONNECTED              26698    /run/systemd/journal/syslog  
unix    8      [ ]                 DGRAM                  CONNECTED              26700    /run/systemd/journal/dev-log  
unix    3      [ ]                 STREAM                 CONNECTED              51389    /run/systemd/journal/stdout
```

37. Command: netstat -r

Interpretation: Display the kernel routing table.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ netstat -r  
Kernel IP routing table  
Destination      Gateway           Genmask           Flags      MSS Window  irtt Iface  
default          NEXUS.mshome.ne  0.0.0.0           UG         0 0        0 eth0  
172.28.160.0     0.0.0.0          255.255.240.0     U          0 0        0 eth0  
samir@NEXUS:~$
```

38. Command: netstat -p

Interpretation: Show the PID and name of the program to which each socket belongs.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ netstat -p  
(Not all processes could be identified, non-owned process info  
will not be shown, you would have to be root to see it all.)  
Active Internet connections (w/o servers)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Prog  
ram name  
tcp        0      0 172.28.164.27:44116    carpaccio.canonical:https ESTABLISHED -  
  
Active UNIX domain sockets (w/o servers)  
Proto RefCnt Flags               Type           State         I-Node  PID/Program name      Path  
unix  2      [ ]                 DGRAM          -            -         /var/run  
/chrony/chronyd.sock  
unix  2      [ ]                 DGRAM          29895       355/systemd           /run/use  
n/1000/systemd/notify
```

39. Command: sudo tcpdump

Interpretation: Print a description of the contents of packets on a network (requires superuser privileges).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo tcpdump  
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode  
listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes  
00:25:29.607592 IP carpaccio.canonical.com.https > 172.28.164.27.44116: Flags [.], seq 8  
38410027:838411427, ack 2115694335, win 503, options [nop,nop,TS val 3790868718 ecr 1718  
286262], length 1400  
00:25:29.659194 IP carpaccio.canonical.com.https > 172.28.164.27.44116: Flags [.], seq 1  
400:2800, ack 1, win 503, options [nop,nop,TS val 3790868768 ecr 1718286317], length 140  
0  
00:25:29.659372 IP 172.28.164.27.44116 > carpaccio.canonical.com.https: Flags [.], ack 2  
800, win 7799, options [nop,nop,TS val 1718286651 ecr 3790868718], length 0  
00:25:29.669725 IP carpaccio.canonical.com.https > 172.28.164.27.44116: Flags [.], seq 2  
800:4200, ack 1, win 503, options [nop,nop,TS val 3790868779 ecr 1718286317], length 140  
^  
960 packets captured  
964 packets received by filter  
0 packets dropped by kernel
```

40. Command: host [google.com](https://www.google.com)

Interpretation: Perform a DNS lookup to convert the domain name to an IP address.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ host google.com  
google.com has address 142.250.195.14  
google.com has IPv6 address 2404:6800:4002:818::200e  
google.com mail is handled by 10 smtp.google.com.  
samir@NEXUS:~$
```

41. Command: `tracert` [youtube.com](https://www.youtube.com)

Interpretation: Trace the path to a network host, discovering the MTU (Maximum Transmission Unit) along the way.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ tracert youtube.com  
  1?: [LOCALHOST] pmtu 1500  
  1: NEXUS.mshome.net 0.557ms  
  1: NEXUS.mshome.net 0.183ms  
  2: 192.168.1.254 1.478ms  
  3: 27.34.0.1 9.282ms  
  4: be-83-46.45.gwc-ndc-core-01.wlink.com.np 9.607ms  
  5: ae-20-136.41.gwj-htda-core-01.wlink.com.np 10.477ms  
  6: ae-21-139.41.gwj-btwl-core-01.wlink.com.np 15.206ms  
  7: ae52-ipt-bhwa-01.wlink.com.np 13.782ms  
  8: 103.225.212.15 29.037ms !H  
Resume: pmtu 1500  
samir@NEXUS:~$
```

42. Command: `ifconfig`

Interpretation: Display the status of currently active network interfaces.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ ifconfig  
ipmiutil config ver 3.18  
Cannot map memory.  
Cannot open an IPMI driver: /dev/imb, /dev/ipmi0, /dev/ipmi/0,  
or direct driverless.  
ipmiutil config, cannot open IPMI driver  
samir@NEXUS:~$
```

43. Command: `route`

Interpretation: Show the IP routing table.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ route  
Kernel IP routing table  
Destination      Gateway           Genmask           Flags Metric Ref    Use Iface  
default          NEXUS.mshome.ne  0.0.0.0           UG    0      0      0 eth0  
172.28.160.0     0.0.0.0          255.255.240.0     U      0      0      0 eth0  
samir@NEXUS:~$
```

44. Command: nslookup [google.com](https://www.google.com)

Interpretation: Query an Internet domain name server interactively.

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ nslookup google.com  
Server:          10.255.255.254  
Address:         10.255.255.254#53  
  
Non-authoritative answer:  
Name:   google.com  
Address: 142.250.195.14  
Name:   google.com  
Address: 2404:6800:4002:818::200e  
  
samir@NEXUS:~$
```

45. Command: whois [chat.com](https://www.chat.com)

Interpretation: Guess the right server to ask for the specified object (in this case, the domain [chat.com](https://www.chat.com)).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ whois chat.com  
Domain Name: CHAT.COM  
Registry Domain ID: 721179_DOMAIN_COM-VRSN  
Registrar WHOIS Server: whois.markmonitor.com  
Registrar URL: http://www.markmonitor.com  
Updated Date: 2024-11-13T14:54:45Z  
Creation Date: 1996-09-20T04:00:00Z  
Registry Expiry Date: 2027-09-19T04:00:00Z  
Registrar: MarkMonitor Inc.  
Registrar IANA ID: 292  
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com  
Registrar Abuse Contact Phone: +1.2086851750  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
```

Package Management Commands

46. Command: sudo apt-get update

Interpretation: Resynchronize the package index files from their sources (requires superuser privileges).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo apt-get update  
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease  
Hit:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease  
Hit:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease  
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease  
Reading package lists... Done  
samir@NEXUS:~$
```

47. Command: sudo apt-get upgrade

Interpretation: Install the newest versions of all packages currently installed on the system from the sources enumerated in /etc/apt/sources.list (requires superuser privileges).

Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo apt-get upgrade  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
Calculating upgrade... Done  
The following packages will be upgraded:  
  distro-info-data nano python3-urllib3 snapd vim vim-common vim-runtime vim-tiny xxd  
9 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
Need to get 36.2 MB of archives.  
After this operation, 2134 kB of additional disk space will be used.  
Do you want to continue? [Y/n] y  
Get:1 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 distro-info-data all 0.5  
2ubuntu0.8 [5302 B]  
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates/main amd64 vim amd64 2:8.2.3995-1ub
```

48. Command: sudo apt-get install sl

Interpretation: Install the package 'sl' (requires superuser privileges).

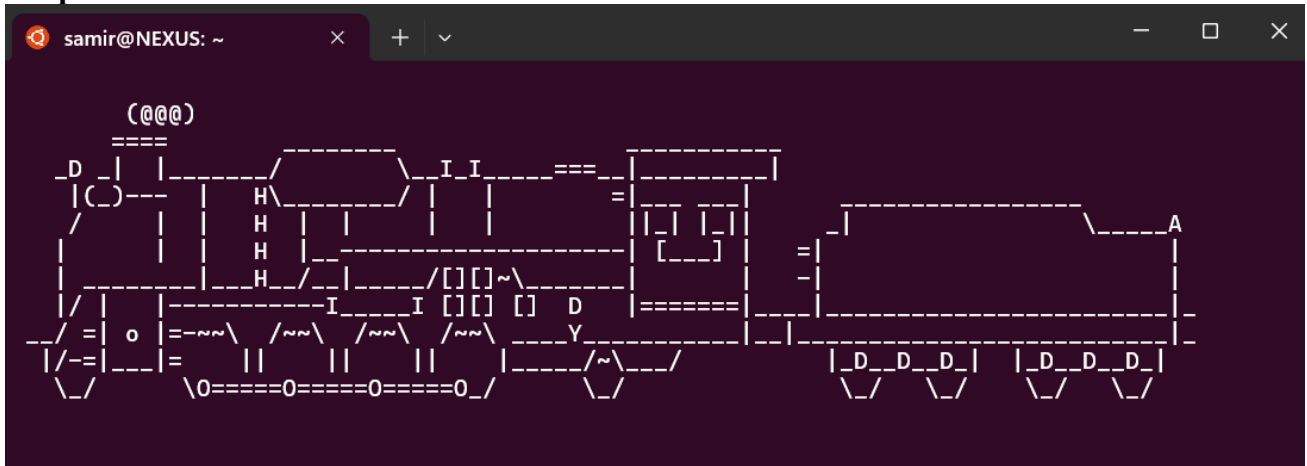
Output:

```
samir@NEXUS: ~  
samir@NEXUS:~$ sudo apt-get install sl  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
The following NEW packages will be installed:  
  sl  
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.  
Need to get 12.7 kB of archives.  
After this operation, 60.4 kB of additional disk space will be used.  
Get:1 http://archive.ubuntu.com/ubuntu jammy/universe amd64 sl amd64 5.02-1 [12.7 kB]  
Fetched 12.7 kB in 0s (29.6 kB/s)  
debconf: DbDriver "config": /var/cache/debconf/config.dat is locked by another process:  
Resource temporarily unavailable  
Selecting previously unselected package sl.
```

49. Command: sl

Interpretation: Display animations aimed to correct users who accidentally enter 'sl' instead of 'ls'.

Output:



50. Command: exit

Interpretation: Exit the current program or shell.